

PSI-FL Relation Table

Abstract protocols and a keyword-attention instantiation

Relation-table construction runs once before federated learning. We first specify five abstract algorithms: exact PSI, description-based relation construction, fuzzy PSI, affinity-based relation construction, and an image filter. We then instantiate the semantic representation with keyword attention and describe the implemented precision-first extension. The first five algorithms deliberately leave encoders, affinity functions, and cryptographic mechanisms unspecified.

Notation. Client i holds local classes C_i and a private map π_i from local ids to masked ids. An external class handle is $(i, \pi_i(a))$; raw local ids and private class samples stay local. A relation is a pair of these handles, optionally with a matching weight. $\text{Mutual}(R)$ selects positive, unique maxima in both their row and column; zero scores and ties abstain. All candidate matrices below use masked-id order.

Abstraction boundary. Exact PSI tests equality of canonical description tokens. Fuzzy PSI tests a public similarity predicate. Neither primitive alone establishes that two classes have the same semantic identity. Algorithms 1 and 2 specify a conceptual exact-PSI baseline, not an existing `run_rt_protocol` branch. The remaining abstractions organize the fuzzy and image-filtering behavior used by the implementation. An ideal primitive’s stated output is not a claim that the current CKKS prototype achieves that primitive’s ideal leakage guarantee.

I. Abstract protocols

Algorithm 1 Simple exact PSI (abstract primitive)

Require: client i ’s private set S_i ; client j ’s private set S_j

Ensure: common tokens I to the designated participants

- 1: **procedure** EXACTPSI(S_i, S_j)
- 2: jointly evaluate set equality through an exact-PSI primitive
- 3: $I \leftarrow S_i \cap S_j$
- 4: **return** I
- 5: **end procedure**

This is a functionality specification: neither party sends its plaintext set to the other. A selected implementation must define its security model and leakage (for example set sizes and the intersection).

Algorithm 2 Description-based relation construction using exact PSI

Require: local classes C_i , descriptions $d_i[a]$, agreed canonicalization Canon

Ensure: masked-id relations E

```
1: for each client  $i$  do
2:    $u_i[a] \leftarrow \text{Canon}(d_i[a])$  for every  $a \in C_i$ 
3:   retain  $(a, \pi_i(a), u_i[a])$  in a private local table
4: end for
5:  $E \leftarrow \emptyset$ 
6: for each pair  $i < j$  do
7:    $I \leftarrow \text{EXACTPSI}(\{u_i[a]\}_a, \{u_j[b]\}_b)$  // Algorithm 1
8:   for each  $u \in I$  do
9:     exchange masked handles associated with  $u$ 
10:    if  $u$  identifies exactly one class  $a$  at  $i$  and one class  $b$  at  $j$  then
11:      add  $(i, \pi_i(a), j, \pi_j(b), 1)$  to  $E$ 
12:    end if
13:  end for
14: end for
15: return  $E$  for global relation-table construction
```

Equality is on the *whole canonical description*, not merely one shared keyword. A common vocabulary alone does not ensure identical descriptions. Synonyms and different languages require a shared canonicalization or the fuzzy extension below. Ambiguous duplicate descriptions are left unresolved. Releasing matched masked handles is an extra relation-disclosure step beyond plain set intersection.

Algorithm 3 Labeled fuzzy PSI with a threshold ladder (abstract primitive)

Require: private labeled objects $\{x_a\}_a, \{y_b\}_b$; public similarity sim

Require: public thresholds Δ and allowed-pair mask M

Ensure: pairwise ranks R indexed by masked handles

```
1: procedure FPSIRANK( $\{x_a\}, \{y_b\}, \text{sim}, \Delta, M$ )
2:   jointly evaluate the following functionality without exchanging raw objects:
3:   for each pair  $(a, b)$  do
4:      $R[a, b] \leftarrow M[a, b] \sum_{\delta \in \Delta} \mathbf{1}\{\text{sim}(x_a, y_b) > \delta\}$ 
5:   end for
6:   return  $R$  to the designated participants
7: end procedure
```

A positive rank nominates a similar pair; more crossed thresholds indicate stronger evidence. The strict inequality defines a similarity threshold, not a distance tolerance. This labeled variant exposes pair identities and quantized similarity ranks; it is more informative than an unlabeled fuzzy intersection. The abstract loop specifies an output, not plaintext evaluation by a trusted server. Cryptographic realization is deferred to Part II.

Algorithm 4 Affinity-based relation construction using fuzzy PSI

Require: local class objects $x_i[a]$; public ordered references $P = (p_1, \dots, p_m)$

Require: affinity $\text{aff}(x, y)$, profile transform $\mathcal{N}$, profile similarity sim , Δ

Ensure: weighted masked-id relations E and optional rank matrices

```
1: for each client  $i$  and each  $a \in C_i$  do
2:    $s_i[a] \leftarrow \mathcal{N}([\text{aff}(x_i[a], p_1), \dots, \text{aff}(x_i[a], p_m)])$ 
3: end for
4:  $E \leftarrow \emptyset$ 
5: for each client pair  $i < j$  do
6:    $R_{ij} \leftarrow \text{FPSIRANK}(s_i, s_j, \text{sim}, \Delta, \mathbf{1})$ 
7:   for each  $(a, b) \in \text{MUTUAL}(R_{ij})$  do
8:     add  $(i, \pi_i(a), j, \pi_j(b), R_{ij}[a, b])$  to  $E$ 
9:   end for
10: end for
11: return  $E$  and  $\{R_{ij}\}$ 
```

// Algorithm 3

$\text{aff}(x, y)$ expresses a class object's affinity to a public reference. Fuzzy PSI compares the resulting *profiles* through sim ; these are distinct operations. No concrete encoder, cosine formula, or classifier is assumed here. The reference order and profile coordinates must be compatible across clients. Image-based profiles instantiate **affscan**; text attention provides another shared-reference construction.

Algorithm 5 Image filter for candidate relations (abstract cascade)

Require: candidate pairs E'_{ij} from a preceding description/semantic stage

Require: local image-class objects X_i^a , public image references P_I

Require: image affinity aff_I , transform $\mathcal{N}_I$, similarity sim_I , Δ_I

Ensure: surviving candidate pairs E_{ij} and image ranks R_I

```
1: for each client  $i$  and class  $a$  do
2:    $s_i^I[a] \leftarrow \mathcal{N}_I([\text{aff}_I(X_i^a, p)]_{p \in P_I})$ 
3: end for
4: for each client pair  $i < j$  do
5:    $A \leftarrow \{a : (a, b) \in E'_{ij}\}; B \leftarrow \{b : (a, b) \in E'_{ij}\}$ 
6:    $M[a, b] \leftarrow \mathbf{1}\{a \in A \text{ and } b \in B\}$ 
7:    $R_I \leftarrow \text{FPSIRANK}(s_i^I, s_j^I, \text{sim}_I, \Delta_I, M)$  // Algorithm 3
8:    $E_{ij} \leftarrow E'_{ij} \cap \text{MUTUAL}(R_I)$ 
9: end for
10: return  $\{E_{ij}, R_I\}_{i < j}$ 
```

This is an intersection filter: image evidence can remove candidates but cannot recover a pair discarded by the preceding stage. The mask covers the Cartesian product of candidate endpoint sets, so image rivals are compared within those sets. The preceding stage may be Algorithm 2, Algorithm 4, or keyword attention. In code, `filter` uses direct-description fuzzy ranks before this filter; `attn_filter/intersection` uses attention-derived fuzzy ranks instead.

Assembling the global table (abstract). Collect weighted masked relations, process stronger edges first, and merge compatible components. Never put two classes of the same client in one component. Assign each component a global id and retain unmatched classes as singletons. For precision-first matching, also require support for every cross-component pair, as detailed after Algorithm 7. Subsequent FL rounds query each client by its masked handle; the client resolves the local id privately.

Reading order. Algorithms 1–2 establish exact equality and its relation-table use. Algorithm 3 replaces equality with a similarity predicate; Algorithm 4 supplies affinity profiles to it. Algorithm 5 adds visual corroboration. The following section makes the semantic and visual representations concrete, while retaining Algorithm 3 as the abstract fuzzy-PSI interface.

II. Implemented keyword attention and precision-first matching

Keyword-level input. A description is a list of concepts, such as `[digit, zero]`. Each keyword is encoded separately; the list is never joined into a sentence before encoding. All clients share the same ordered public text-anchor list A , while each client embeds it using its own text encoder E_i . The current implementation uses one head and no learned query/key projection. Local inverse document frequency (IDF), rather than a learned keyword gate, downweights words repeated across local classes.

Conventions for the concrete algorithms. UNIT normalizes each row with denominator $\max(\|v\|_2, 10^{-12})$. $\text{FPSIRANK}(U, V, \Delta, M)$ abbreviates Algorithm 3 with $\text{sim}(u, v) = u^\top v$ on normalized coordinates. MUTUAL retains positive unique row-and-column maxima.

Algorithm 6 Keyword-attention representation and standalone matching

Require: ordered public text anchors A , client-local text encoder E_i , keyword lists $D_i[a]$, $\tau > 0$

Ensure: text coordinates $z_i[a]$ indexed by the public anchor list

```

1: for each client  $i$  do
2:    $K_i \leftarrow \text{UNIT}([E_i(w)]_{w \in A}); n \leftarrow |C_i|$ 
3:    $K_{c,i} \leftarrow K_i - \text{columnmean}(K_i)$ 
4:   compute compact SVD  $K_{c,i} = U \Sigma V^\top$ 
5:   retain singular values  $\sigma > \max(10^{-6} \sigma_{\max}, 10^{-12})$ 
6:    $W_i \leftarrow U_{\text{keep}} \Sigma_{\text{keep}}^{-1} U_{\text{keep}}^\top$ 
7:    $\text{df}_i(w) \leftarrow |\{a \in C_i : w \in D_i[a]\}|$ 
8:   for each class  $a \in C_i$  do
9:     encode every keyword separately:  $q_k \leftarrow \text{UNIT}(E_i(D_i[a]_k))$ 
10:     $c_a(w) \leftarrow$  multiplicity of keyword  $w$  in  $D_i[a]$ 
11:     $\tilde{w}_k \leftarrow \log(n/\text{df}_i(D_i[a]_k))/c_a(D_i[a]_k)$ 
12:    if all  $\tilde{w}_k = 0$  then
13:       $\tilde{w}_k \leftarrow 1/c_a(D_i[a]_k)$  for every keyword
14:    end if
15:     $w_k \leftarrow \tilde{w}_k / \sum_\ell \tilde{w}_\ell$ 
16:     $\alpha_k \leftarrow \text{softmax}(q_k K_i^\top / \tau)$  for each keyword
17:     $\bar{\alpha}_i[a] \leftarrow \sum_k w_k \alpha_k$ 
18:     $\ell_i[a] \leftarrow \log(\max(\bar{\alpha}_i[a], \varepsilon_{\text{tiny}}))$  elementwise
19:     $z_i[a] \leftarrow \text{UNIT}((\ell_i[a] - \text{mean}(\ell_i[a]))W_i)$ 
20:  end for
21: end for
Standalone attn matching.
22: for each pair  $i < j$  do
23:    $R_T \leftarrow \text{FPSIRANK}(z_i, z_j, \Delta_T, \mathbf{1})$ 
24:    $R_T[a, b] \leftarrow 0$  wherever  $R_T[a, b] < |\Delta_T|$ 
25:   emit masked relations from  $\text{MUTUAL}(R_T)$ , weighted by  $R_T[a, b]$ 
26: end for
```

IMPLEMENTATION. `keyword_attn` and `anchor_log_coordinates`, default `coord="log_whiten"`, one head. With legacy `desc_vecs`, a whole description is one query and receives weight 1; without keyword metadata, weights are uniform. No projection head or keyword weights are trained. Different text encoders embed the same anchor *list*; identical numerical keys are not required. This shared indexing does not guarantee semantic alignment of arbitrary encoders. With one active keyword, the corrected coordinate is an anchor-span projection.

Concrete visual signals and fuzzy-PSI realization

Let F be the common frozen image encoder and P_I the ordered public, unlabeled image probes. Define

$$\mu = |P_I|^{-1} \sum_{p \in P_I} F(p), \quad r_i(a) = |X_i^a|^{-1} \sum_{x \in X_i^a} F(x) - \mu, \quad h_p = F(p) - \mu.$$

The concrete instantiation of the abstract image affinity is

$$\begin{aligned} \text{aff}_I(X_i^a, p) &= \text{UNIT}(r_i(a))^\top \text{UNIT}(h_p). \\ s_i[a] &= \text{UNIT}([\text{aff}_I(X_i^a, p)]_p - \text{mean}_p \text{aff}_I(X_i^a, p)). \end{aligned}$$

Thus profile matching compares $s_i[a]^\top s_j[b]$. Precision-first adds full-feature vectors $v_i[a] = \text{UNIT}(r_i(a))$ and compares $v_i[a]^\top v_j[b]$ as a separate verification signal. These visual signals share the same encoder and samples, so they are not statistically independent evidence.

Data use. The MNIST harness uses a shared pretrained ResNet-18 with its classifier head removed. Legacy summaries use at most 64 samples/class; precision mode uses `precision_summ` from all available samples, falling back to `summ` if unavailable. Server setting `rt_exemplars_per_class=0` selects all samples. Eligible classes must meet the configured sample minimum; ineligible classes remain singleton ids. No cross-client ground-truth labels decide matches.

Fuzzy PSI realization. For Algorithm 3, the CKKS receiver encrypts its coordinate vectors; the sender homomorphically computes dot products and returns encrypted values $(u^\top v - \delta)rM[a, b]$, with independent positive $r \sim \text{Uniform}(1, 1000)$. The receiver decrypts, applies the strict positive test (and mask), sums threshold bits, and returns ranks. This prototype reveals masked magnitudes to the receiver as well as ranks and relevant sizes/masks; it is not a proven realization of a bits-only ideal functionality. Approximation can alter bits near thresholds. `psi="plain"` is a non-private reference evaluation.

Rank bounds for verification. For sorted thresholds $d_1 \leq \dots \leq d_m$, set $d_0 = -1$ and $d_{m+1} = 1$. For rank r , $\text{RANKBOUNDS}(r, \Delta) = (d_r, d_{r+1})$ gives conservative lower and upper bounds for evaluated, unmasked pairs with exact threshold bits. Mask-forced zeros give no similarity bound. CKKS numerical errors mean these are not unconditional certificates in encrypted execution. The experimental ladders are

$$\begin{aligned} \Delta_T &= \{.90, .85, \dots, .40\}, \quad \Delta_A = \{.990, .985, \dots, .950\}, \\ \Delta_V &= \{.50, .55, \dots, .85\} \cup \{.900, .902, \dots, .998\}. \end{aligned}$$

These are configurable settings, not guarantees of semantic correctness.

Algorithm 7 Precision-first attention-affinity preprocessing: current `psi_attn_filter`

Require: z_i from Algorithm 6, (s_i, v_i) from the concrete visual construction

Require: ladders $\Delta_T, \Delta_A, \Delta_V$; verification floor $\eta = .90$, margin $\gamma = .01$

Ensure: weighted edges E , supported pair set H , uncertain candidates for local review

```
1:  $E, H \leftarrow \emptyset$ 
2: for each client pair  $i < j$  do
3:    $R_T \leftarrow \text{FPSIRANK}(z_i, z_j, \Delta_T, \mathbf{1})$ 
4:    $M[a, b] \leftarrow \mathbf{1}\{R_T[a, b] > 0\}$ 
5:    $R_A \leftarrow \text{FPSIRANK}(s_i, s_j, \Delta_A, M)$  // zero if no allowed pairs
6:    $R_V \leftarrow \text{FPSIRANK}(v_i, v_j, \Delta_V, \mathbf{1})$  // all competitors, unmasked
7:    $(L_T, U_T) \leftarrow \text{RANKBOUNDS}(R_T, \Delta_T)$ 
8:    $(L_A, U_A) \leftarrow \text{RANKBOUNDS}(R_A, \Delta_A)$ ;  $(L_V, U_V) \leftarrow \text{RANKBOUNDS}(R_V, \Delta_V)$ 
9:    $B \leftarrow 0$ ;  $G \leftarrow 0$ 
10:  for each pair  $(a, b)$  do
11:     $T \leftarrow [L_T[a, b] \geq \max \Delta_T]$ ;  $I \leftarrow [L_A[a, b] \geq \max \Delta_A]$ 
12:    if  $R_T[a, b] = 0$  or  $R_A[a, b] = 0$  or  $\neg(T \vee I)$  or  $L_V[a, b] < \eta$  then
13:      continue
14:    end if
15:     $q \leftarrow \max(\{U_V[a, b'] : b' \neq b\} \cup \{U_V[a', b] : a' \neq a\} \cup \{-1\})$ 
16:    if  $(T \wedge L_V[a, b] \geq q - \gamma)$  or  $(\neg T \wedge L_V[a, b] \geq q + \gamma)$  then
17:       $B[a, b] \leftarrow 1$ ;  $G[a, b] \leftarrow (L_T[a, b] + L_A[a, b] + L_V[a, b])/3$ 
18:      add the masked unordered pair  $\{(i, \pi_i(a)), (j, \pi_j(b))\}$  to  $H$ 
19:    end if
20:  end for
21:  add  $(i, \pi_i(a), j, \pi_j(b), G[a, b])$  to  $E$  for  $(a, b) \in \text{MUTUAL}(G)$ 
22:  record strongly nominated pairs failing verification as uncertain, not as confirmed errors
23: end for
24: return  $(E, H)$  for the consistent global grouping described below; retain review records locally
```

IMPLEMENTATION. `attn_match="precision"`, `verified_candidates`, and `rank_bounds`. Confidence and margins use only threshold ranks. R_V must include every image-feature rival: hiding rivals before checking the margin could create artificial confidence. Strong text may resolve visually close alternatives within γ ; weak-text image rescue instead requires a margin of γ over all rivals. The image-feature floor applies in both cases.

What precision-first changes (and what it preserves)

Both standalone `attn` and precision-first `attn_filter` use Algorithm 6’s keyword queries, local IDF weights, public text anchors, and corrected log-attention coordinates. Precision-first is a *matching and verification policy*; it does not train a stronger attention encoder.

Aspect	Original standalone <code>attn</code>	Precision-first <code>attn_filter</code>
Semantic signal	Keyword-attention coordinates	Same keyword-attention coordinates
Pair nomination	Text must cross the highest text threshold	Positive text and profile ranks; at least one crosses its highest threshold
Visual evidence	None	Image profiles plus full-feature verification
Rival comparison	Unique mutual maximum of text ranks	Full-feature bounds against all row/column rivals, then mutual fused score
Local summaries	No images used	All available samples when supplied
Global grouping	Same-client exclusion	Same-client exclusion plus complete-link support
Uncertainty	Unmatched classes remain separate	Also records verification and grouping conflicts for review

The two earlier filtered variants. The legacy `intersection` variant runs text mutual matching first and then Algorithm 5; it can only discard earlier candidates. The `fusion` variant retains positive text/profile ranks, requires at least one highest-rung rank, and performs mutual matching on $R_T|\Delta_A| + R_A|\Delta_T|$. Neither has the extra full-feature verification or complete-link requirement. Precision-first can rescue a weak but positive text match using strong image profile evidence, provided the full-feature verification beats every rival by the required margin. It cannot rescue a zero-text-rank pair.

Strong-text versus image-led verification. Every accepted pair must satisfy $L_V \geq .90$ by default. Let q be the largest full-feature upper bound among every alternative in the pair’s row and column. Strong-text candidates require $L_V \geq q - .01$: text may resolve a visually close ambiguity. Weak-text, strong-profile candidates instead require $L_V \geq q + .01$. Thus the strong-text case permits a tolerance; it does not demand a strict visual winning margin. Ranking then averages text, profile, and verification lower bounds, using only fuzzy-PSI outputs.

Consistent global grouping. Sort selected edges by score. Merge two components only if they have no client in common and every cross-component pair belongs to the supported set H from Algorithm 7. A transitive path alone is insufficient. Rejected merges are recorded; remaining singletons retain their own global ids. Enforcing this rule discloses supported masked-pair relations H to the aggregator in addition to selected edges.

Accuracy objective and cost. The goal is fewer false merges, accepting extra offline computation, communication, and potentially lower recall. All-sample summaries, a finer verification ladder, and all-rival comparisons increase cost. Sparse or ambiguous data may still prevent correct mappings, and no perfect-mapping guarantee follows from these checks. The centrally orchestrated experiment simulates client roles; a deployment must implement their communication and account for the stated leakage. The training adapter `to_group_map` uses a known shared party/local-label schema and adds no new semantic evidence.